

CH9.2 對數與對數函數

$$1.1 < x < 3, \text{ 但 } x \neq 2 \quad 2. \frac{4}{3} \quad 3.6 \quad 4.2 \quad \text{或} \quad \log_3 144 \quad 5.1 \quad 6.2\sqrt{3} + 2$$

$$7.1 < x < 3 \quad 8.54\sqrt{3} \quad 9. \frac{1}{12} \quad 10.4 \quad 11.196800 \quad 12.16 \quad 13.33 \quad 14.48 \quad 15.2$$

解析

1. $\log_{(x-1)}(4x - x^2 - 3)$ 有意義

$$\Rightarrow \begin{cases} x-1 > 0 \\ x-1 \neq 1 \\ 4x-x^2-3 > 0 \end{cases} \Rightarrow \begin{cases} x > 1 \\ x \neq 2 \\ 1 < x < 3 \end{cases}$$

$$\therefore 1 < x < 3, \text{ 但 } x \neq 2$$

2. $9^{\log_2 2} + 2^{-\log_4 9} - 5^{\frac{\log 3}{\log 5}} = (3^2)^{\log_2 2} + (2^{\log_2 3^2})^{-1} - 5^{\log_3 3}$

$$= (2)^2 + (3)^{-1} - 3 = 4 + \frac{1}{3} - 3 = \frac{4}{3}$$

3. $\log_2 3 \times \log_3 5 \times \log_5 7 \times \log_7 64$

$$= \frac{\log 3}{\log 2} \times \frac{\log 5}{\log 3} \times \frac{\log 7}{\log 5} \times \frac{\log 64}{\log 7} = \frac{\log 2^6}{\log 2} = \underline{6}$$

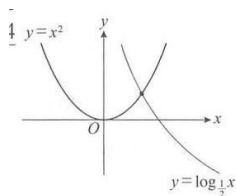
4. $\log_3(3^x + 36) = \frac{x}{2} + 1 + \log_3 5 = \log_3 3^{\frac{x}{2}} + \log_3 3 + \log_3 5$

$$= \log_3 \left(3^{\frac{x}{2}} \cdot 3 \cdot 5 \right)$$

$$\therefore 3^x + 36 = 15 \left(3^{\frac{x}{2}} \right)$$

$$\Leftrightarrow \left(3^{\frac{x}{2}} \right)^2 - 15 \left(3^{\frac{x}{2}} \right) + 36 = 0 \Leftrightarrow \left(3^{\frac{x}{2}} - 3 \right) \left(3^{\frac{x}{2}} - 12 \right) = 0$$

$$\Leftrightarrow 3^{\frac{x}{2}} = 3 \quad \text{或} \quad 3^{\frac{x}{2}} = 12 \Leftrightarrow \frac{x}{2} = 1 \quad \text{或} \quad \frac{x}{2} = \log_3 12$$



$$\Rightarrow x = \underline{2} \text{ 或 } x = 2\log_3 12 = \log_3 144, y = x^2$$

$$5. \begin{cases} y = x^2 \\ y = \log_{\frac{1}{2}} x \end{cases}$$

之圖形如右，有 1 個交點

$$6. \begin{cases} x = k \\ y = \log_3 x \end{cases} \Leftrightarrow A(k, \log_3 k)$$

$$\begin{cases} x = k \\ y = \log_3(x + 4) \end{cases} \Rightarrow B(k, \log_3(k + 4))$$

$$\overline{AB} = \log_3(k + 4) - \log_3 k = \log_3 \frac{k + 4}{k} = \frac{1}{2} = \log_3 3^{\frac{1}{2}}$$

$$\Rightarrow \frac{k + 4}{k} = \sqrt{3} \Rightarrow k + 4 = \sqrt{3}k$$

$$\therefore k = \frac{4}{\sqrt{3} - 1} = \underline{2\sqrt{3} + 2}$$

$$7. 1 + \log_{\frac{1}{2}}(x - 1) > \log_{\frac{1}{4}}(4 - x)$$

$$\Rightarrow \log_{\frac{1}{4}} \frac{1}{4} + \log_{\frac{1}{4}}(x - 1)^2 > \log_{\frac{1}{4}}(4 - x)$$

$$\Rightarrow \log_{\frac{1}{4}} \frac{1}{4} (x - 1)^2 > \log_{\frac{1}{4}}(4 - x)$$

$$\Rightarrow \frac{1}{4} (x^2 - 2x + 1) < (4 - x) \Rightarrow x^2 - 2x + 1 < 16 - 4x$$

$$\Rightarrow x^2 + 2x - 15 < 0 \Rightarrow (x + 5)(x - 3) < 0 \Rightarrow -5 < x < 3$$

但須滿足 $x - 1 > 0$ 且 $4 - x > 0 \therefore 1 < x < 3$

$$8. \log_3 x + \log_3 y = \log_3 xy = 6 = \log_3 3^6 \therefore xy = 3^6 = 729$$

$$x + 3y \geq 2\sqrt{x \cdot 3y} = 2\sqrt{3 \cdot 729} = 2 \cdot 27\sqrt{3} = 54\sqrt{3}$$

$\therefore$ 最小值為 $\underline{54\sqrt{3}}$

$$9. \text{ 令 } x = \sqrt{4 + \sqrt{7}} - \sqrt{4 - \sqrt{7}}$$

$$\begin{aligned} \Rightarrow x^2 &= (4 + \sqrt{7}) - 2\left(\sqrt{4 + \sqrt{7}}\right)\left(\sqrt{4 - \sqrt{7}}\right) + (4 - \sqrt{7}) = 8 - 2\sqrt{16 - 7} \\ &= 8 - 2 \times 3 = 2 \end{aligned}$$

$$\therefore x = \sqrt{2}$$

$$\log_{64} \left(\sqrt{4 + \sqrt{7}} - \sqrt{4 - \sqrt{7}} \right) = \log_{2^6} \sqrt{2} = \frac{1}{6} \log_2 2 = \underline{\underline{\frac{1}{12}}}$$

$$10. \because x > 1$$

$$2\log_3 x + \log_x 9 = 2\log_3 x + 2\log_x 3 \geq 2\sqrt{2\log_3 x \cdot 2\log_x 3} = 2 \times 2 = 4$$

$\therefore$ 最小值為 4

$$11. \because \log 0.1968 \text{ 之尾數為 } \log 1.968, \log 378600 \text{ 之首數為 } 5 \therefore \log x = 5 + \log 1.968 = \log 196800, \text{ 即 } x = 196800$$

$$12. 100^n > 2^{100} \Rightarrow \log 100^n > \log 2^{100} \Leftrightarrow n \times 2 > 100 \times \log 2 \approx 30.10 \Rightarrow n > 15.05, \text{ 即 } n \geq 16$$

$\therefore$ 最小正整數為 16

$$13. \log 2^{106} = 106 \times \log 2 \approx 106 \times 0.3010 = 31.906$$

其首數為 31，為 32 位數

$$\text{且 } \log 8 \approx 0.9030 < 0.906 < 0.9542 \approx \log 9$$

最高位數字為 8

$$\log 3^{66} = 66 \times \log 3 \approx 66 \times 0.4771 = 31.4886$$

其首數為 31，為 32 位數

$$\text{且 } \log 3 \approx 0.4771 < 0.4886 < 0.6020 \approx \log 4$$

最高位數字為 3

$\therefore 2^{106} + 3^{66}$ 為 33 位數字

14. 令本金為 a , 需時 n 年

$$a \times (1 + 2.4\%)^n > 3a$$

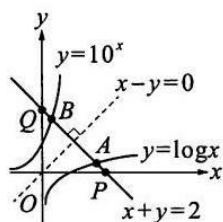
$$\Leftrightarrow (1.024)^n > 3 \Rightarrow \log(1.024)^n > \log 3$$

$$\Rightarrow n(\log 1024 - 3) > \log 3$$

又 $\log 1024 = \log 2^{10} = 10 \log 2 \approx 3.01$

$$\Rightarrow n(3.01 - 3) > 0.4771 \Rightarrow n > \frac{0.4771}{0.01}$$

$\Leftrightarrow n > 47.71 \Leftrightarrow n$ 至少為 48 年



15. 由於 $y = 10^x$ 與 $y = \log x$ 兩圖形對稱於直線 $x - y = 0$

又直線 $x + y = 2$ 分別交 x 軸、 y 軸於 $P(2, 0)$ 、 $Q(0, 2)$

且 P 、 Q 兩點對稱於直線 $x - y = 0$ 令 $A(x_1, y_1), B(x_2, y_2)$

則 $\overline{AB}$ 的中點就是 $\overline{PQ}$ 的中點，其坐標為 $(1, 1)$

即 $\frac{x_1 + x_2}{2} = 1$, 故 $x_1 + x_2 = 2$